

Greater Houston Metro Market Evaluation

Grease Trap Waste Disposal Demand & Competitive Landscape Analysis

Prepared for the platform
the Managing Director

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Executive Summary

The Greater Houston metropolitan area represents a compelling market opportunity for the platform Infrastructure's pretreatment facilities. With 14,532 food service establishments generating grease trap waste across a 9-county MSA, Houston is the fourth-largest food service market in the United States. Despite this scale, the disposal infrastructure is severely concentrated: all 10 registered grease trap waste (5GG) processing facilities are located within Harris County, leaving 8 of 9 MSA counties with zero GTW processing infrastructure.

The automated gap analysis identified 54 high-density restaurant zones, of which 27 (50%) are classified as underserved or moderately gapped, with the nearest disposal facility more than 10 miles away. Fort Bend County (1,522 food establishments, zero 5GG facilities), Montgomery County (1,199 establishments, zero facilities), and Galveston County (780 establishments, zero facilities) emerge as primary market-entry targets. Regulatory tailwinds further strengthen the opportunity: Houston's December 2023 tightening of FOG discharge limits from 400 mg/L to 200 mg/L effectively doubles surcharge exposure for high-FOG dischargers, creating immediate new demand for pretreatment.

1. Market Overview

The Houston-The Woodlands-Sugar Land Metropolitan Statistical Area (MSA) spans 9 counties with an estimated population of 7.3 million residents, making it the fifth-largest MSA in the United States. The region's economy supports a substantial food service industry, with 14,532 food service and drinking place establishments (NAICS 722) according to the 2023 U.S. Census County Business Patterns.

Harris County, home to the City of Houston, dominates the MSA with 10,038 food establishments representing 69.1% of the total market. Fort Bend County (1,522 establishments, 10.5%), Montgomery County (1,199, 8.2%), Galveston County (780, 5.4%), and Brazoria County (689, 4.7%) comprise the next tier of significant markets. The remaining four counties (Chambers, Liberty, Waller, and Austin) collectively represent less than 2.5% of the MSA total.

Houston operates one of the largest separate sewer systems in the country, with 39 wastewater treatment plants, 6,100+ miles of sewer pipe, and 383 lift stations. The city is currently under a federal EPA Consent Decree for Clean Water Act violations, committing \$3-5 billion over 22 years to address sanitary sewer overflows and effluent limit exceedances. This regulatory environment creates significant tailwinds for pretreatment infrastructure.

Food Service Establishments by County

County	Food Establishments	% of MSA	5GG Facilities
Harris	10,038	69.1%	9
Fort Bend	1,522	10.5%	0
Montgomery	1,199	8.2%	0
Galveston	780	5.4%	0
Brazoria	689	4.7%	0
Chambers	80	0.6%	0
Liberty	102	0.7%	0
Waller	65	0.4%	0
Austin	57	0.4%	1
Total	14,532	100%	10

Source: U.S. Census Bureau, County Business Patterns, 2023. NAICS 722 (Food Services and Drinking Places). 5GG facility count from TCEQ MSW Facility Registry.
Data: [Houston GTW Competitors.xlsx](#)

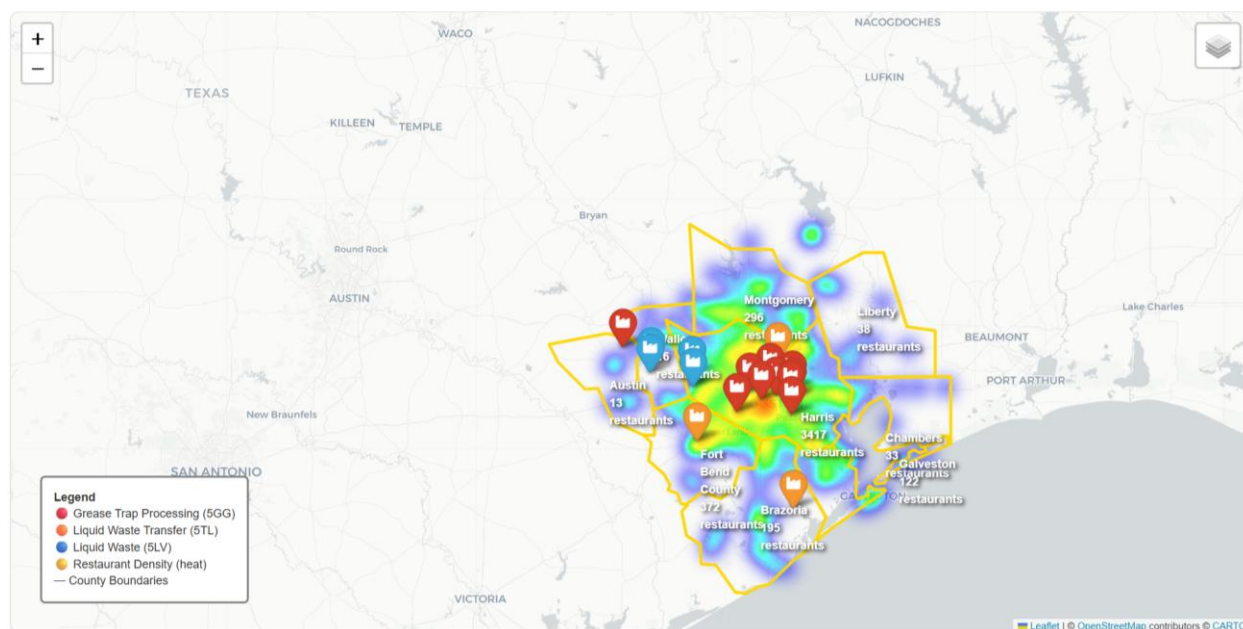
2. Demand Analysis

To identify areas of concentrated restaurant density, 4,502 food establishment locations were geocoded across the 9-county MSA using OpenStreetMap data. While this represents approximately 31% of the Census total, the spatial density pattern is representative, as OpenStreetMap coverage is proportionally consistent across counties.

Key Density Nodes

The heat map analysis reveals concentrated restaurant clusters in the following areas. In Harris County, the highest density is found in Central Houston (Downtown/Midtown, Galleria/Uptown, and Westheimer corridor), with secondary nodes in Chinatown/Bellaire, the Heights, and West University Place. Significant suburban density exists in Katy, Humble/Atascocita, Cypress, and Spring/Klein.

Outside Harris County, notable density clusters include The Woodlands and Conroe in Montgomery County, Sugar Land and Rosenberg in Fort Bend County, the Seawall/61st Street corridor in Galveston, and Lake Jackson/Angleton in Brazoria County. These suburban and exurban nodes are experiencing rapid population growth, suggesting future restaurant density increases.



Map: [Houston Market Map.html](http://HoustonMarketMap.html)

3. Competitive Landscape

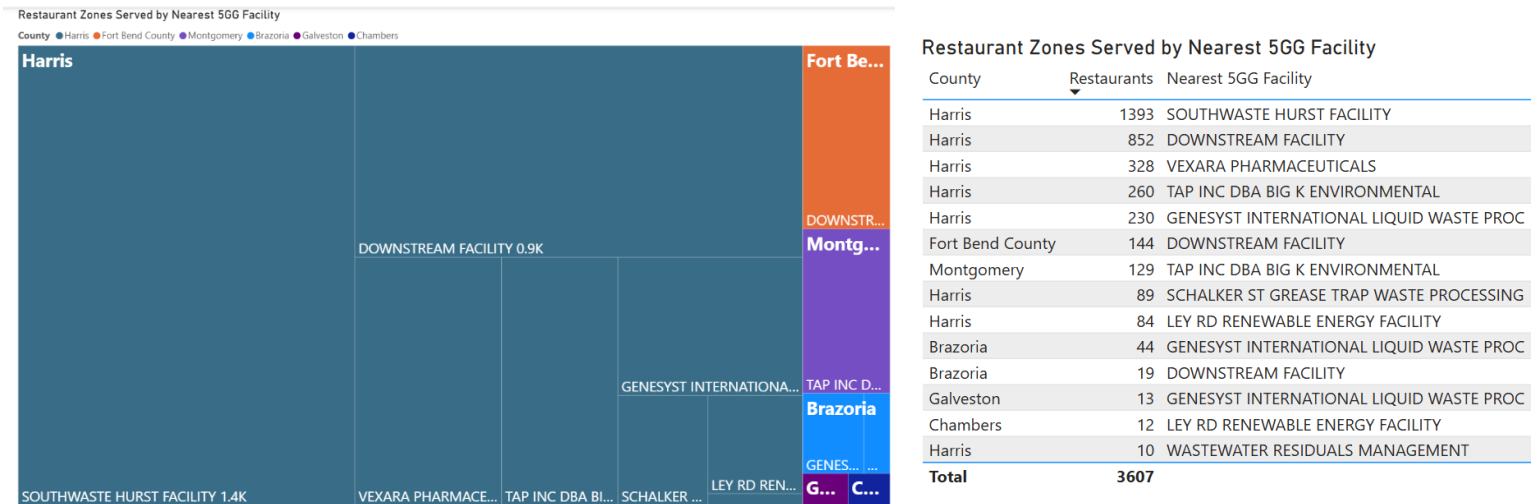
Using the Texas Commission on Environmental Quality (TCEQ) Municipal Solid Waste Facility Registry, 22 active liquid waste handling facilities were identified across the Houston MSA. Of these, 10 are classified as 5GG (grease trap waste processing), which represents the platform's direct competitive set. The remaining facilities include 5TL (liquid waste transfer stations) and 5LV (general liquid waste facilities).

Competitor Concentration

All 9 out of 10 registered 5GG facilities are located within Harris County and 1 is in Austin pending legal status. Zero grease trap processing facilities exist in any other MSA county. This represents a stark infrastructure imbalance: 8 of 9 counties, collectively home to 4,494 food service establishments (31% of the MSA total), have no registered GTW processing infrastructure.

Within Harris County, competitor facilities are concentrated in central and north Houston. Southwaste Hurst Facility, established in 1948 near the Katy area, serves as the nearest disposal option for the largest number of restaurant zones (1,393 restaurants across its service area). Downstream Facility serves 852 restaurant zones, while Vexara Pharmaceuticals, TAP Inc (Big K Environmental), and Genesyst International Liquid Waste Processing serve the remaining central Houston market.

Notably, even within Harris County where all 10 facilities are located, the restaurant-to-facility ratio is approximately 1,004:1 — indicating that existing infrastructure is already stretched thin. Suburban Harris County corridors including Katy, Humble, Cypress, and Spring face disposal distances of 10-20 miles despite being in the same county as the facilities.



Market Sizing: Estimated GTW Volume and Revenue Potential

Southwaste Disposal, the dominant operator in Harris County, publicly reports processing 275 million+ gallons of liquid waste in 2025, with a 2026 target of 302.5 million gallons (source: southwaste.com). Southwaste is a subsidiary of WRM Company and represents the single largest GTW processing operation in the Houston MSA.

This volume data, combined with publicly available surcharge rates, enables a preliminary estimate of the economic value at stake in the Houston GTW market.

At Houston's current surcharge rate of \$2.50–\$5.00 per 1,000 gallons above domestic baseline, the surcharge exposure on 275 million gallons of high-strength waste represents approximately \$687,500–\$1,375,000 in annual surcharge costs that would be incurred if this waste were discharged directly to the municipal sewer without pretreatment. This establishes the economic floor, the minimum value that pretreatment saves the market annually, from a single operator alone.

From the platform's revenue perspective, industry tip fees for grease trap waste disposal typically range from \$0.05–\$0.15 per gallon. Applied to Southwaste's 275 million-gallon throughput, this implies annual tip fee revenue of \$13.75 million–\$41.25 million for a single dominant operator. The fact that Southwaste is actively growing and its 2026 target of 302.5 million gallons represents a 10% year-over-year increase, which confirms that disposal demand is expanding, not contracting.

Critically, Southwaste's entire operation is concentrated in Harris County. The 4,494 food establishments in the remaining 8 MSA counties generate grease trap waste that must either travel to Harris County for processing or find alternative disposal, both of which represent addressable demand for a new pretreatment facility. Even capturing 15% of Southwaste's current throughput volume in underserved suburban markets would represent 27.5–41.25 million gallons annually, translating to estimated tip fee revenue of \$1.4–\$6.2 million per facility.

4. Gap Analysis

An automated spatial gap analysis was conducted to quantify the disposal infrastructure deficit. The Houston MSA was divided into a grid of approximately 2km x 2km cells. Cells in the top 15% by restaurant count were classified as high-density zones and clustered into 54 distinct restaurant concentration zones. For each zone, the distance to the nearest 5GG facility was calculated using the haversine formula.

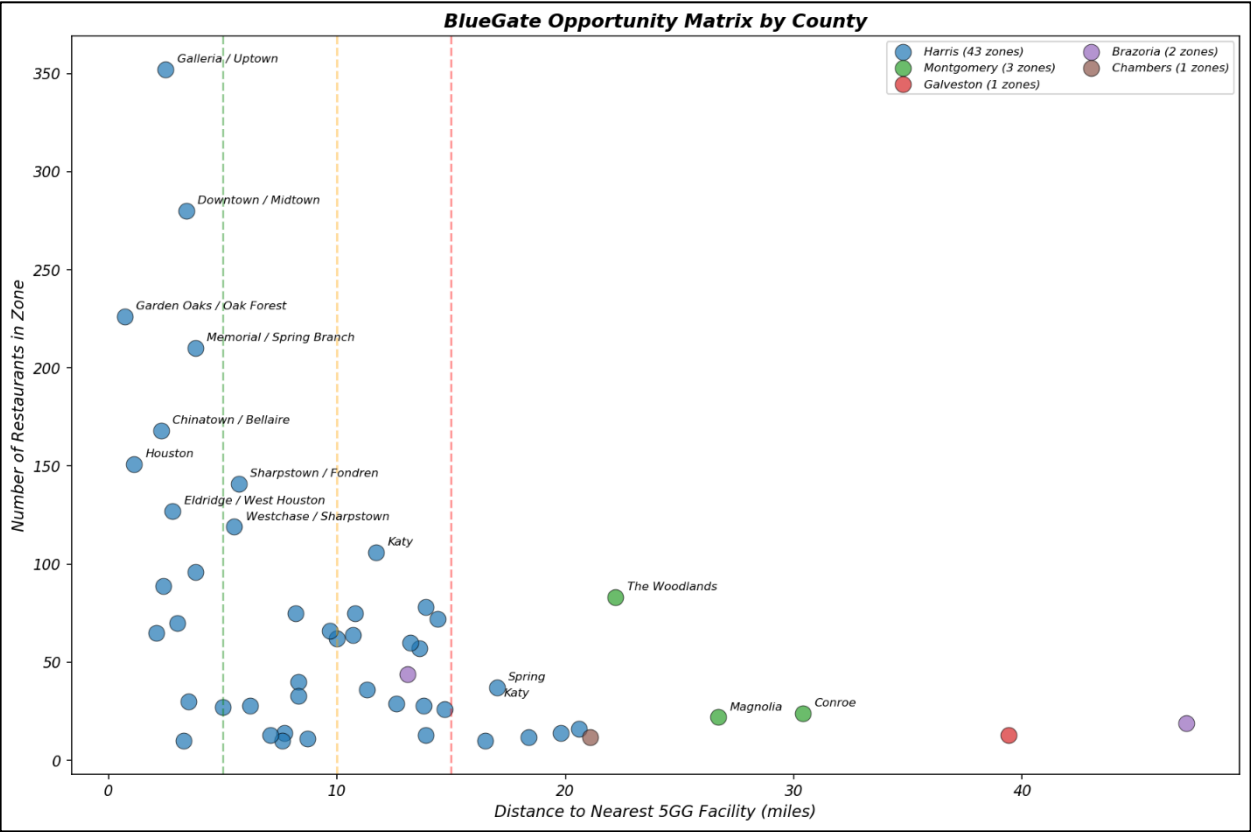
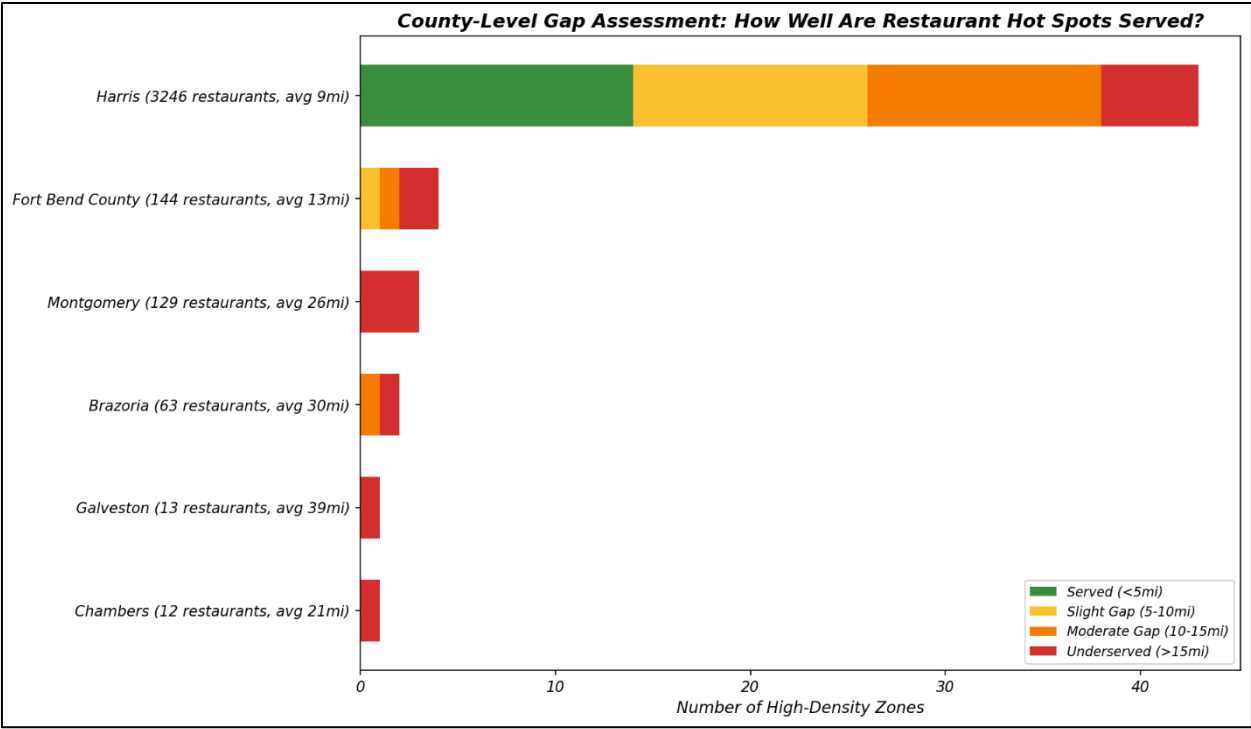
Gap Assessment Summary

Classification	Zones	Restaurants	Avg Distance
UNDERSERVED (>15 mi)	13	301	23.4 mi
MODERATE GAP (10-15 mi)	14	700	12.3 mi
SLIGHT GAP (5-10 mi)	13	661	7.8 mi
SERVED (<5 mi)	14	1,945	2.5 mi
Total	54	3,607	11.7 mi

50% of high-density restaurant zones (27 of 54) are more than 10 miles from the nearest 5GG facility, indicating haulers in those areas face significant travel distances for disposal. The average distance across all zones is 11.7 miles, suggesting systemic underinvestment in disposal infrastructure relative to demand distribution.

County-Level Findings

The gap analysis reveals a clear core-periphery pattern. Central Houston zones cluster in the SERVED category (nearest facility under 5 miles), while every zone outside the central core faces increasing disposal distances. All Montgomery County zones are UNDERSERVED (22-30 miles). All Fort Bend County zones except Sugar Land are UNDERSERVED or MODERATE GAP. Galveston County's single zone is 39.4 miles from the nearest facility



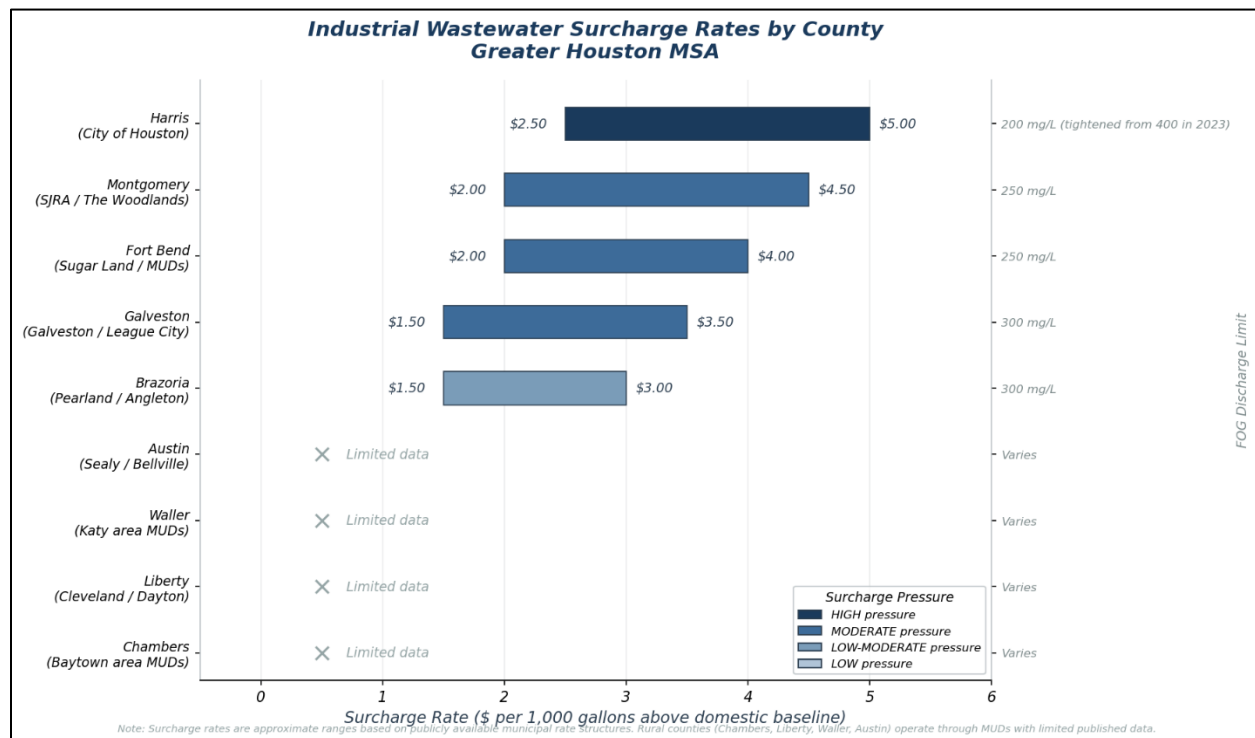
5. Regulatory Tailwinds

The regulatory environment in the Houston MSA creates significant demand-side tailwinds for pretreatment infrastructure.

FOG Discharge Limit Tightening

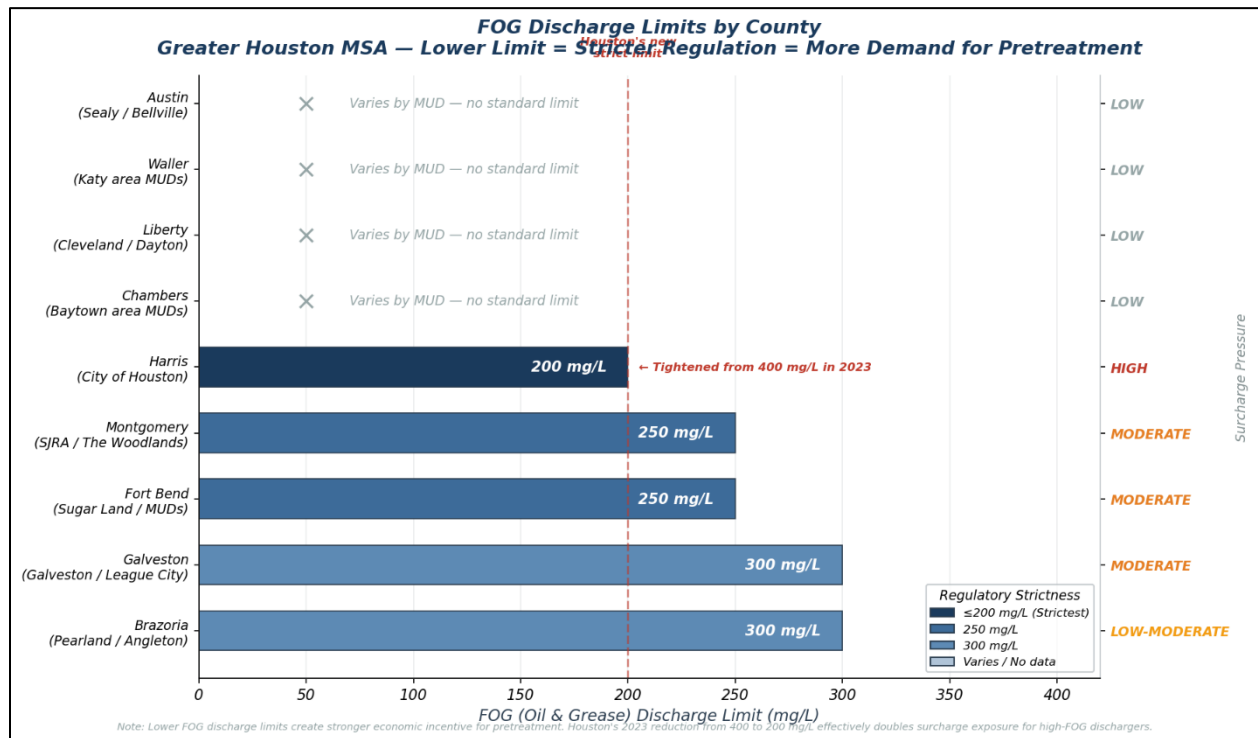
In December 2023, the City of Houston reduced its FOG (oil and grease) discharge limit from 400 mg/L to 200 mg/L over a two-year phase-in period. This is the most significant regulatory change affecting the Houston GTW market. Food service establishments that were previously compliant at 350 mg/L are now in violation and face surcharge exposure. This effectively expands the platform's addressable market within Harris County— restaurants and food processors that previously had no economic incentive for pretreatment now face material cost pressure without it.

Fort Bend and Montgomery counties maintain 250 mg/L limits, while Galveston and Brazoria apply 300 mg/L thresholds. As municipalities face increasing FOG-related infrastructure strain, further limit reductions across the MSA are probable, creating a secular growth driver for pretreatment demand.



Surcharge Rates

Harris County commands the highest industrial wastewater surcharge rates in the MSA, with formula-based rates ranging from approximately \$2.50 to \$5.00 per 1,000 gallons above domestic baseline. Montgomery County follows at \$2.00-\$4.50. These surcharge levels create a direct economic incentive for pretreatment: it is cheaper for food service establishments and haulers to pretreat FOG upstream than to absorb escalating surcharges on high-strength discharge.



EPA Consent Decree

Houston's EPA Consent Decree requires the city to invest \$3-5 billion over 22 years to address chronic sewer overflows and treatment plant violations. This regulatory pressure incentivizes the city to welcome pretreatment infrastructure that reduces FOG and solids loading on the municipal sewer system. The platform's facilities, which discharge only pretreated effluent under controlled permit conditions, align with the city's compliance objectives.

6. Opportunity Zones & Recommendations

Primary Opportunity: Fort Bend County (Sugar Land / Rosenberg)

Fort Bend County presents the strongest immediate market-entry opportunity. With 1,522 food service establishments (Census 2023), a population of approximately 870,000, and zero 5GG grease trap processing facilities, the county represents a significant disposal infrastructure void. The only waste-related facility in Fort Bend is a single 5TL (liquid waste transfer station), which does not provide the pretreatment processing that the platform offers. Haulers serving Sugar Land, Rosenberg, and Missouri City currently drive 10-17 miles into Harris County for disposal. Fort Bend's surcharge rates (\$2.00-\$4.00/kgal) and 250 mg/L FOG limit create moderate but growing economic pressure for pretreatment. A pretreatment facility positioned near Sugar Land would capture significant hauler volume from this rapidly growing suburban market.

Secondary Opportunity: Montgomery County (The Woodlands / Conroe)

Montgomery County offers a compelling secondary target. With 1,199 food establishments, a population exceeding 650,000 (one of the fastest-growing counties in Texas), and zero 5GG facilities, the disposal gap is absolute. The analysis identified 296 mapped restaurant locations, with all high-density zones classified as UNDERSERVED, averaging 26 miles from the nearest processing facility. The Woodlands alone, with 83 mapped restaurants in a single high-density zone, is 22.2 miles from the nearest 5GG facility (TAP Inc in north Harris County). The San Jacinto River Authority manages wastewater services with a 250 mg/L FOG limit and surcharge rates of \$2.00-\$4.50/kgal. A pretreatment facility in south Montgomery County would serve a large, growing, and completely unserved market.

Tertiary Opportunity: Harris County Suburban Ring (Katy / Humble / Cypress)

While Harris County is home to all 10 existing 5GG facilities, those facilities are concentrated in central Houston. The county's suburban ring, particularly Katy (106+ restaurants, 11.7 miles from nearest facility), Humble/Atascocita (57+ restaurants, 13.6 miles), and Cypress (13.9 miles), faces disposal distances comparable to other underserved counties. Notably, Southwaste Hurst Facility, established in 1948, has captured the central Houston market with 1,393 restaurant zones in its service area. However, with Houston's FOG discharge limit now tightened to 200 mg/L, the demand for pretreatment across the entire county is increasing. The suburban corridors represent an opportunity for a new entrant to serve the growing outer ring that existing central facilities cannot efficiently reach.

Emerging Market: Brazoria County

Brazoria County should not be overlooked despite its smaller market size. With 689 food establishments, a population of approximately 419,000 that has grown significantly since 2020, and zero 5GG facilities, it represents an emerging opportunity. The county's two high-density zones (Pearland and Lake Jackson area) contain 63 mapped restaurants and are 13-47 miles from the nearest disposal facility. Surcharge rates of \$1.50-\$3.00/kgal, while lower than Harris County, still create economic incentive for pretreatment. As the platform is in its early platform-building stage, capturing smaller but completely uncontested markets builds operational track record and revenue base alongside larger market entries. A 5TL liquid waste transfer facility exists in Brazoria, but no 5GG pretreatment facility, indicating demand for waste handling exists, but full pretreatment capacity does not.

7. Data Sources & Methodology

Data Sources

- U.S. Census Bureau, County Business Patterns 2023 — NAICS 722 food service establishment counts by county. Official federal data source providing the demand baseline of 14,532 establishments. [Data_Census.csv](#)
- OpenStreetMap via Overpass Turbo — 4,502 geocoded food establishment locations across 9 counties. Used for spatial density analysis and heat map generation. Coverage represents ~31% of Census total; spatial density patterns are proportionally representative. [Houston_All_Restaurants.csv](#)
- TCEQ Municipal Solid Waste Facility Registry — 22 active liquid waste facilities in the Houston MSA (10 classified as 5GG grease trap waste processing). Official Texas state regulatory database providing the competitive landscape. [Houston_GTW_Competitors.xlsx](#)
- EPA ECHO Database — NPDES permit data for wastewater treatment plant capacity verification.
- City of Houston Public Works FOG Program — FOG discharge limits and surcharge rate structures.

Analytical Methodology

Restaurant location data was collected from OpenStreetMap using the Overpass Turbo API, querying for nodes tagged as restaurant, fast_food, cafe, and food_court across each MSA county. Data was cleaned in Python (pandas) — removing entries with missing coordinates, standardizing county assignments, and deduplicating. Competitor facility data was extracted from the TCEQ MSW Facility Registry, filtered to the 9 MSA counties and facility types 5GG, 5TL, and 5LV. Facilities with missing or invalid coordinates were identified and excluded from spatial analysis.

The automated gap analysis divided the MSA into a 2km x 2km grid, counted restaurants per cell, identified the top 15% of cells by density, and clustered adjacent high-density cells into zones. Distance from each zone center to every 5GG facility was calculated using the haversine formula. Zones were classified as UNDERSERVED (>15 mi), MODERATE GAP (10-15 mi), SLIGHT GAP (5-10 mi), or SERVED (<5 mi).

Tools

Python (pandas, folium, matplotlib) for data processing, spatial analysis, and visualization. Folium for interactive heat map generation. Power BI for treemap and tabular dashboards. OpenStreetMap Overpass Turbo for geocoded restaurant data extraction.

Limitations & Further Research

OpenStreetMap captures approximately 31% of Census-reported food service establishments. While spatial density patterns are representative, production-level analysis would benefit from supplementing with official state and county health department licensing databases for comprehensive coverage. Surcharge rates are approximate ranges based on publicly available municipal rate structures; a detailed normalization model across all target municipalities would strengthen market-entry pricing analysis. Hauler density data, tip fee benchmarks, and facility utilization rates were not publicly available and would require primary research for production-level diligence.

Attachments

- [Houston Market Map.html](#) — Interactive heat map with county boundaries and competitor markers (open in browser)
- [Houston Gap Analysis Complete.csv](#) — Full gap analysis data with zone identification
- [Houston All Restaurants.csv](#) — 4,502 geocoded restaurant locations
- [Houston GTW Competitors.xlsx](#) — TCEQ competitor facility data